Cao Donghao

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Github:https://cdhspace.github.io/

EDUCATIONAL BACKGROUND

Beijing University of Chemical Technology Beijing, China
B.Eng. in Mechanical Design, Manufacturing and Automation Sept. 2022 – June 2026
Main Courses: *Mechanical Drawing, Theoretical Mechanics, Mechanical Principles, Mechanical Design, Mechanical Manufacturing Technology, Hydraulic & Pneumatic Transmission, Control Engineering*

Imperial College London London, UK
Msc in Control and Optimization Sept. 2026 – Nov. 2027
Main Courses: *Control Theory, Optimisation Theory, Linear Systems, System Identification, Model Predictive Control, Convex Optimisation, Nonlinear Control, Machine Learning for Dynamical Systems*

WORK EXPERIENCES

Institute of Mechanical and Electronic Engineering, Tsinghua University Beijing,China
Research Assistant Sept.2024-April 2025

- Implemented Smith predictor-sliding mode control with PSO algorithm on Panasonic A5 electric cylinder system using dSPACE platform, significantly reducing overshoot and settling time
- Tested the operating voltage range of the electric cylinder, optimized its tracking performance, and reduced the delay to 0.2 seconds

Ingersoll-Rand (China) Industrial Equipment Manufacturing Co.,Ltd Wujiang,China
Product R&D Intern Oct.2025-Feb.2026

- Assisted engineers with product prototype research, system identification and control system design to match various equipment parameter standards.
- Tracked R&D progress, communicated technical requirements, sorted out development faults and reported hidden technical risks to product supervisors.

COMPETITIONS / PROJECT EXPERIENCES

Special Agricultural Terrain Path Planning and Vision Picking Robot
Project Leader & Oversea Cooperation Mentor Mar.2024 – Oct. 2025
Funding by *National College Students' innovation outstanding training program* **(X202510010152)**

- Developed intelligent tea bud picking system using improved Mamba-YOLOv8 vision model with custom dataset for crop recognition, designed collaborative robotic arm end-effector, and conducted 3D terrain modeling for complex agricultural terrains
- Long-term collaborated with faculty and students from the School of Computer Science, Universiti Teknologi Malaysia. Served as a mentor for the Southeast Asia Division of the China International College Students' Innovation Competition, conducting cross-border and cross-linguistic joint cooperation.

Home Service Robot Vision System
Vision Team Leader & Hardware Engineer Mar. 2025– May 2025

- Modeled general household scenarios and accomplished a series of motion tasks including guest identification and object handling.
- Won **National First Prize** in **Robocup China** leading computer vision development with YOLOv8 model compression, maintaining 94% object detection accuracy for autonomous household tasks

MORE AWARDS

- ❖ The Chinese Government National Scholarship 2023
- ❖ National Undergraduate Mathematical Contest in Modelling (CUMCM) -**Beijing 1st Honour**
- ❖ 2024 United States Interdisciplinary Subject Modeling Competition **- Honorable Mention**

ADDITIONAL INFORMATION

Language Skills: Chinese (Native), English (IELTS 6.5), PTE(77)
Computer Skills: dSPACE(Advanced), Simulink(Advanced), MATLAB (Advanced), Python (Advanced), C++ (Advanced), SolidWorks (Advanced), AutoCAD (Intermediate), ROS (Intermediate), NX (Intermediate)